

JAVASCRIPT REVIEWER — Lessons 1 & 2

■ What is JavaScript?

JavaScript is a lightweight, object-oriented scripting language that makes web pages interactive. Created in 1995 by Brendan Eich (originally called **Mocha** → **LiveScript** → **JavaScript**). Standardized as **ECMAScript** in 1997. Runs in browsers and (via Node.js) on servers.

HTML = structure | **CSS** = style | **JavaScript** = behavior

■ Syntax Rules

- Statements end with ;
- Code blocks use { }
- Variable names: start with letter, _, or \$
- Case-sensitive (*name* ≠ *Name*)
- No hyphens in names

■ Variable Declaration

Keyword	Use
let	Value may change
const	Value stays fixed
var	Older style (avoid)

■ Operators

Type	Symbols	Example → Result
Arithmetic	+ - * / % ** ++ --	5 + 3 → 8 2**3 → 8 10%3 → 1
Assignment	= += -= *= /= %= **=	x += 3 same as x = x + 3
Comparison	== === != !== > < >= <=	5 === "5" → false 5 == "5" → true
Logical	&& !	true && false → false !true → false
String	+ +=	"Hello" + " World" → "Hello World"
Ternary	condition ? a : b	age>=18 ? "Adult" : "Minor"

■ Data Types

Type	Example
Number	42, 3.14
String	"Hello"
Boolean	true / false
Undefined	declared, no value
Null	intentional empty
Object	{key: value}

■ Comments

```
// Single-line comment
/* Multi-line comment */
```

JAVASCRIPT REVIEWER — Lessons 3, 4 & 5

■ Conditions (Lesson 3)

Used to execute code based on whether a condition is **true** or **false**.

Statement	When to Use
if	One condition
if...else	Two possible outcomes
else if	Multiple conditions/ranges
switch	One var, many exact values
ternary ?:	Short if-else (one line)

if...else example:

```
let score = 70;
if (score >= 75) {
  console.log('Passed');
} else {
  console.log('Failed');
}
```

switch example:

```
switch(day) {
  case 'Mon': ...; break;
  default: ...
}
```

■ Always use **break** in switch to stop fall-through.

■ Loops (Lesson 4)

Loops repeat code until a condition is met.

Loop	Use
for	Known number of times
while	Unknown count, check first
do...while	Run at least once
for...of	Loop array/string values
for...in	Loop object properties

for loop:

```
for (let i=1; i<=5; i++) {
  console.log(i);
}
```

while loop:

```
let i = 1;
while (i <= 5) {
  console.log(i); i++;
}
```

Statement	Effect
break	Stops loop immediately
continue	Skips current iteration
return	Exits function with value

■ Strings (Lesson 5)

Strings store text. Written with double "" or single '' quotes, or backticks `` (template literals).

Operation	Method / Syntax	Example → Output
Concatenation	str1 + str2	"Hi" + " World" → "Hi World"
Length	.length	"JavaScript".length → 10
Uppercase	.toUpperCase()	"hello".toUpperCase() → "HELLO"
Lowercase	.toLowerCase()	"HELLO".toLowerCase() → "hello"
Find text	.includes("word")	"I love JS".includes("JS") → true
Extract	.slice(start, end)	"JavaScript".slice(0,4) → "Java"
Replace	.replace("old","new")	"I like cats".replace("cats","dogs")
Template Literal	`text \${variable}`	`Hello \${name}` → "Hello Arjay"

Control Flow Order: Sequential → Conditional
→ Loop → Jump

Strict equality === checks value AND type

Tip: Use const by default; let when value changes